Job Submission Examples

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Job submission files have the file extension .ssub and are used for submitting multiple jobs or more complex jobs to Slurm. These files give you more options for how you would like your job set up, which are defined in the header. Most of these are optional, however, I would advise using at least the partition, mem, output, error, job-name, mail-user, mail-type, and workdir:

```
#!/bin/bash
#
#SBATCH --partition=<partition name>           //which partition you would like this
#SBATCH --ntasks=<number of tasks>             //define the number of instances of yo
#SBATCH --nodes=<number of nodes>              //define how many compute nodes (max:
#SBATCH --mem=<memory in megabytes>            //define how many megabytes of memory
#SBATCH --time=<time limit>                    //define the limit for how long your j
#SBATCH --output=/<output directory>/%J.stdout  //stdout is where your process writes
#SBATCH --error=/<error directory>/%J.stderr    //stderr is where your process writes
#SBATCH --job-name=<job name>                  //set the name of your job
#SBATCH --mail-user=<your email address>       //if you enter your email address here
#SBATCH --mail-type=ALL                        //specify the circumstances in which S
#SBATCH --chdir=<working directory>             //define where Slurm will execute you
#
#################################################
```

Below this part of the file you simply list out the commands you would like SLURM to run, similar to how you would enter them in the command line. If your script requires any environment modules in order to run, then you would load those here.

Distributed Memory

You might use this in a case where you have multiple processes that need to run a large number of tasks. The allocated memory is then distributed to each node as needed:

```
#!/bin/bash
#
#SBATCH --partition=c3
#SBATCH --ntasks=40
#SBATCH --ntasks-per-node=20
#SBATCH --mem=24000
#SBATCH --output=~/analysis/logs/AA00-preprocessing-%J.stdout
#SBATCH --error=~/analysis/logs/AA00-preprocessing-%J.stderr
#SBATCH --job-name=AA00-preprocessing
#SBATCH --mail-user=my_username@laureateinstitute.org
#SBATCH --mail-type=ALL
#SBATCH --workdir=~/analysis/source/AA00
#
#################################################
module load afni/17.0.09


3dConvolve -arguments
```

Shared Memory

You might use this in a case where you need to use more memory (>12000MB) by sharing across multiple cpu cores on one node:

```
#!/bin/bash
#
#SBATCH --partition=c3
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=2
#SBATCH --nodes=1
#SBATCH --mem=24000
#SBATCH --output=~/analysis/logs/AA00-preprocessing-%J.stdout
#SBATCH --error=~/analysis/logs/AA00-preprocessing-%J.stderr
#SBATCH --job-name=AA00-preprocessing
#SBATCH --mail-user=my_username@laureateinstitute.org
#SBATCH --mail-type=ALL
#SBATCH --workdir=~/analysis/source/AA00
#
#################################################
module load afni/17.0.09


3dConvolve -arguments
```